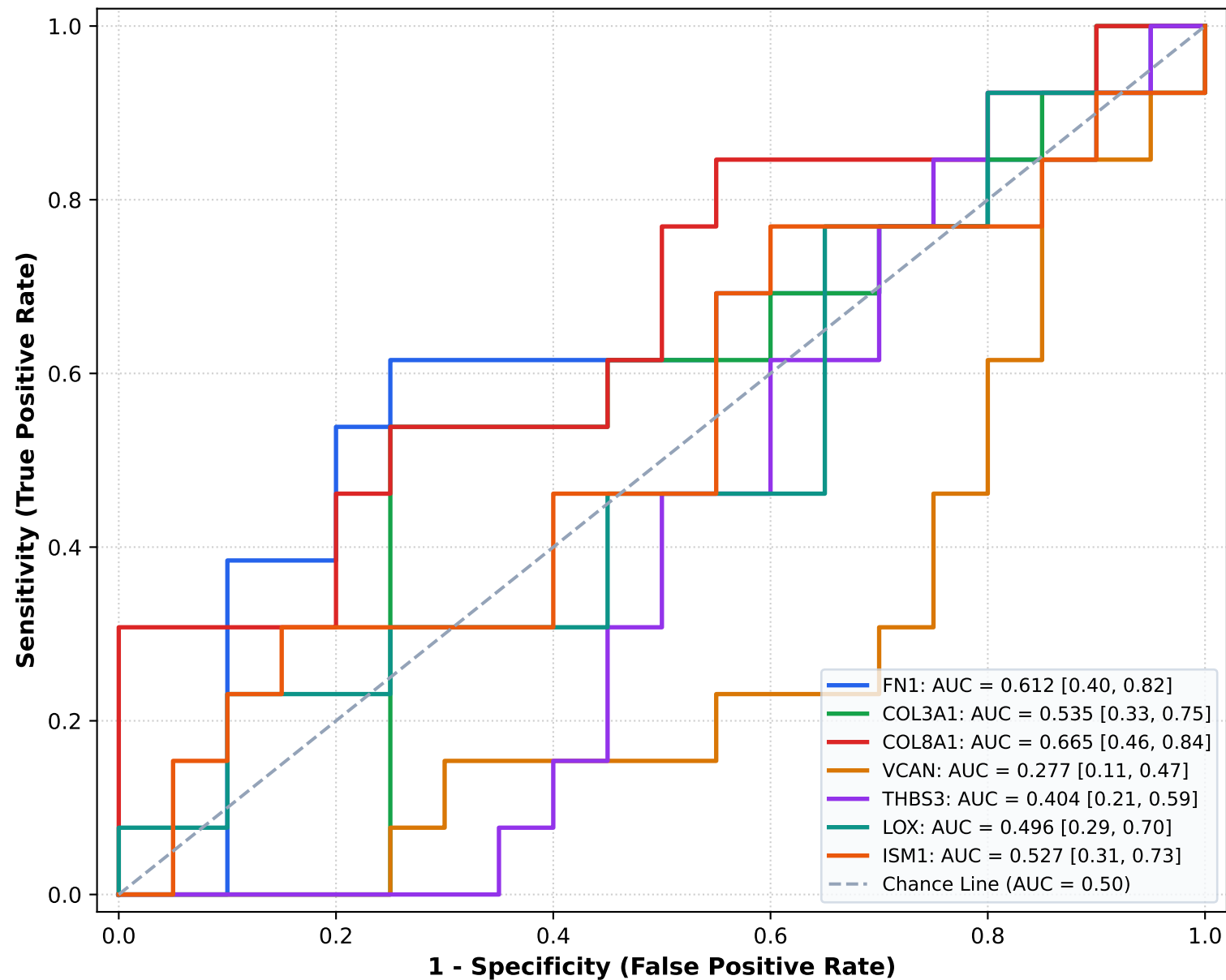


**A. Single-Gene ROC Discrimination in External Effluent Cohort
(GSE125498: N = 33, 13 Late-Stage vs 20 Early-Stage PD)**



**B. Combined Multi-Gene Panel ROC Discrimination
(In-Sample vs 5-Fold Cross-Validation in GSE125498)**

